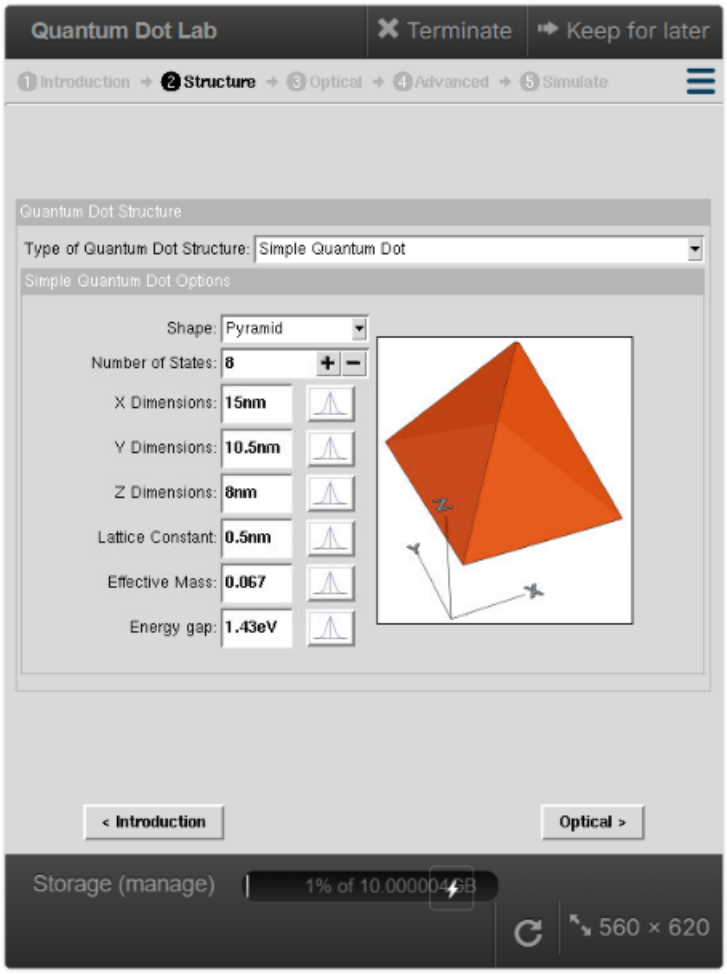
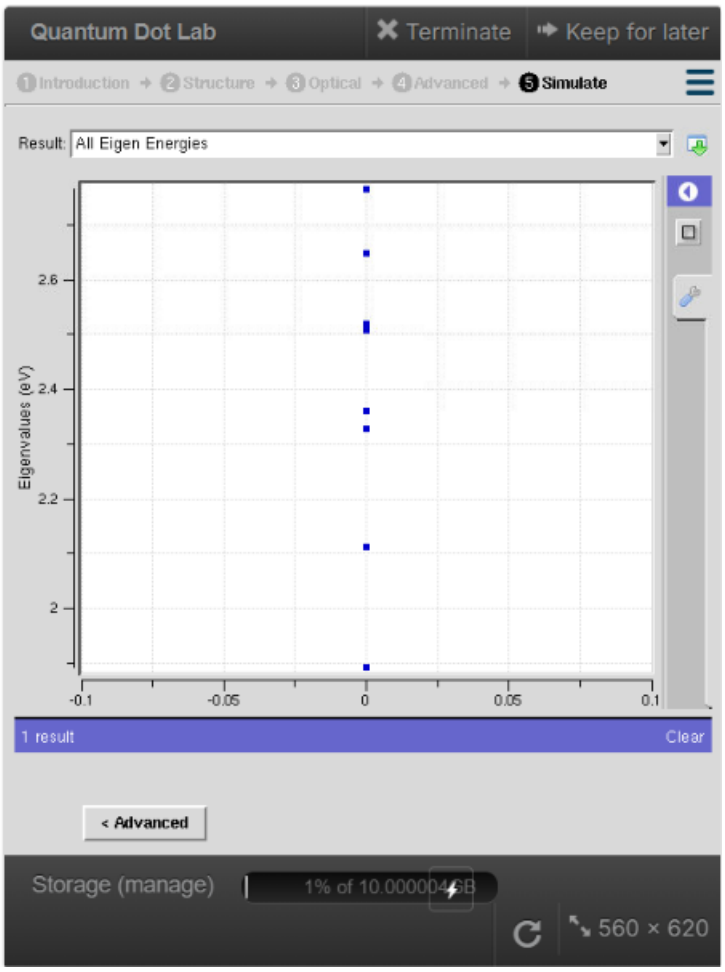
Exp. 3. Simulation of Energy States of Quantum Dot using NanoHub:

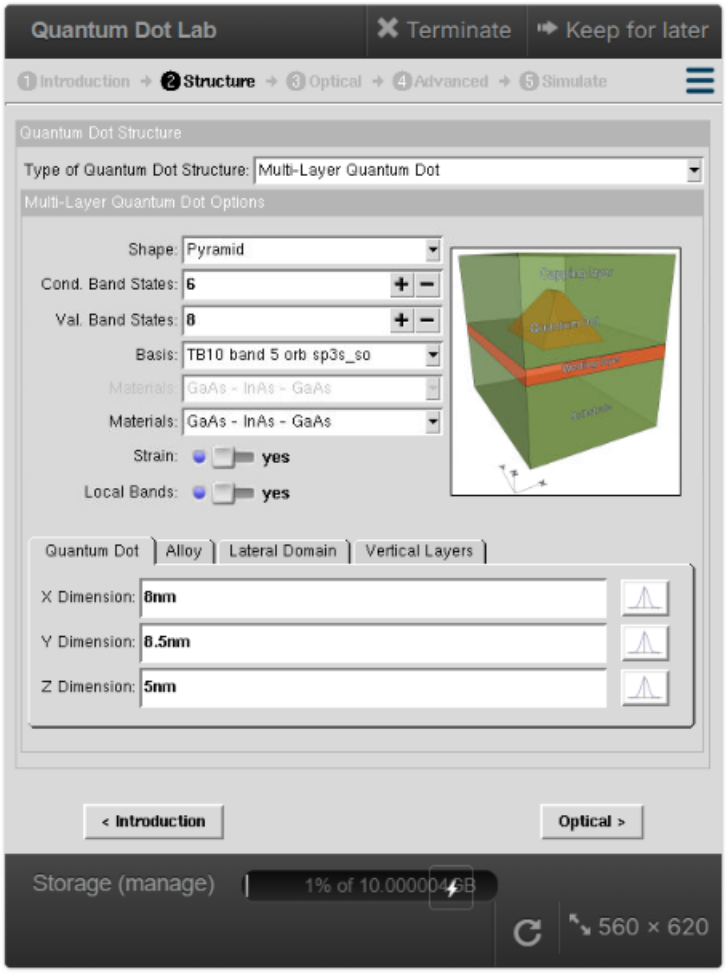
Objective:To obtain the energy states of Quantum dots.

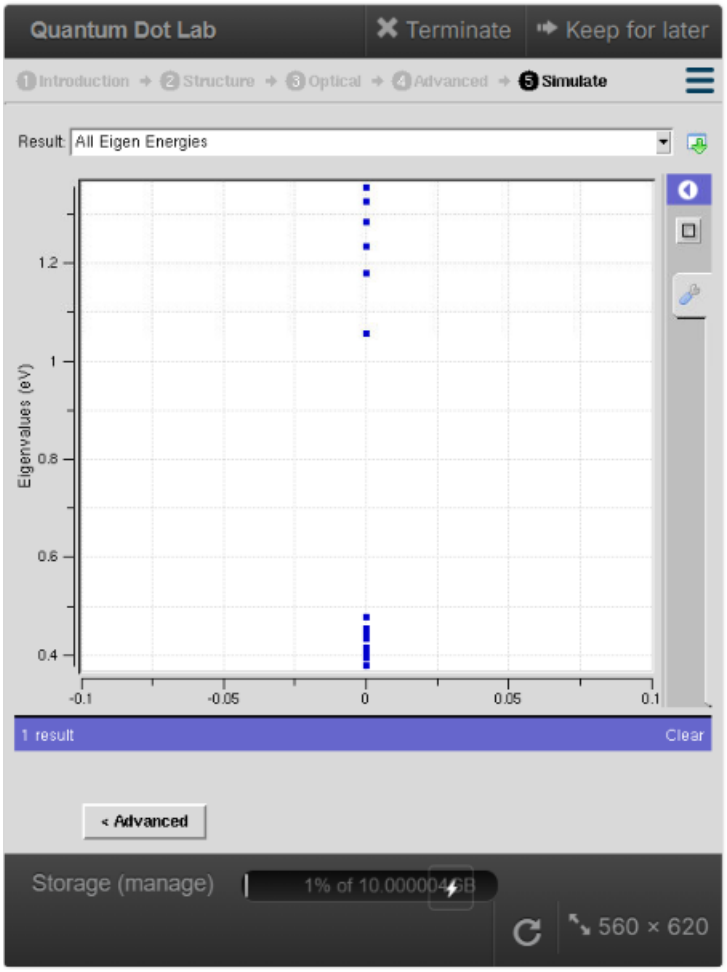
Case Study 1: Simulate the energy states of cylindrical quantum dots of X=15 nm, Y = 10.5nm and Z = 8 nm.





Case Study 2: Simulate the energy states of two layer cuboid quantum dots


Quantum Dot Lab
Terminate
Keep for later
1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate
Quantum Dot Structure
Type of Quantum Dot Structure: Multi-Layer Quantum Dot
Multi-Layer Quantum Dot Options
Shape: Pyramid
Cond. Band States: 6
Val. Band States: 8
Basis: TB10 band 5 orb sp3s_so
Materials: GaAs - InAs - GaAs
Materials: GaAs - InAs - GaAs
Strain: yes
Local Bands: yes
Quantum Dot
Alloy
Lateral Domain
Vertical Layers
X Dimension: 8nm
Y Dimension: 8.5nm
Z Dimension: 5nm
< Introduction
Optical >
Storage (manage)
1% of 10.000004GB
560 × 620



Reflection:

explored three-dimensional quantum confinement by analyzing the discrete energy levels inside various semiconductor shapes. learned that the emission frequencies and electronic properties of quantum dots are heavily size-dependent.
Smaller dot radii yielded larger confinement energies, validating their use in highly tunable optoelectronic devices.